THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Methods of Separation in Everyday Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture of sand and iron in 1:1 by mass weighing 40 g. After using a magnet, the iron obtained should be: 1

(a) 20 g **(b)** 0 g **(c)** 10 g **(d)** 40 g

2. Which is a reason for decanting slowly? 1

(a) to dissolve **(b)** to cool

(c) to avoid disturbing the sediment **(d)** to save time

3. A mixture of water and alcohol cannot be separated by decantation because: 1

(a) water is solid **(b)** alcohol is heavy

(c) alcohol is magnetic **(d)** they mix completely

4. Why would using a very fine sieve not separate a mixture of fine sand and salt? 1

(a) both particles are fine **(b)** sand is soluble

(c) salt is a gas **(d)** sieves are magnetic

5. Which apparatus is used to hold the filter paper while filtering? 1

(a) tripod stand **(b)** funnel **(c)** beaker **(d)** watch glass

6. The method of spinning milk rapidly to separate cream is called: 1

(a) handpicking **(b)** winnowing **(c)** churning **(d)** sieving

7. A mixture of marbles and sand is best separated by: 1

(a) condensation **(b)** sieving **(c)** evaporation **(d)** decantation

8. Which of the following is false about solutions? 1

(a) solutes pass filter paper

(b) they are uniform

(c) they are clear

(d) they scatter light strongly like muddy water

9. **Assertion (A):** Salt dissolved in water can be separated by filtration. 1

Reason (R): The particles of dissolved salt are too small and pass through filter paper.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Sieving separates flour from bran. 1
Reason (R): Sieving works because the particles are of different sizes.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does cream rise to the top of milk kept for some time? 2

12. How can you obtain pure sugar from impure sugar? 2

13. Why does a gold-washing (panning) method use water? 2

14. How can you separate oil from a mixture of oil and water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the speed of dissolving depends on particle size, stirring and temperature. 3

16. How would you know if a liquid is pure water or salt water using evaporation? 3

17. How can you show that a salt solution contains water as well as salt? 3

OR

Design a fair test to find which of salt and sugar is more soluble in water. 3

18. Explain two ways of getting clean water from a muddy puddle in an emergency. 3

OR

How does a vacuum cleaner separate dust from air? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rubina's well water in Raipur looks muddy. Her mother lets it stand in a bucket for an hour. The mud settles at the bottom. She then pours the clear water slowly into another vessel and then passes it through a clean cotton cloth. 5
(i) What is the settling of mud called?
(ii) What is the slow pouring of clear water called?
(iii) Which method uses the cloth?
(iv) Why is the water still not fit to drink even after this? What should be done?

20. Analyse how cream is separated from milk and made into butter at home. 5

OR

Explain why a mixture of sugar and sand is easier to separate in water than by handpicking. 5

END OF QUESTION PAPER — Total: 40 Marks